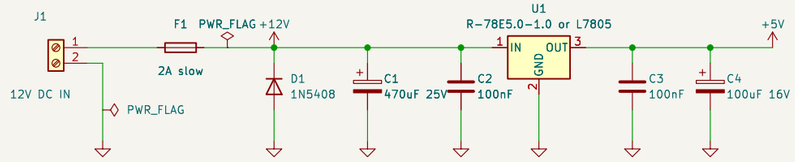
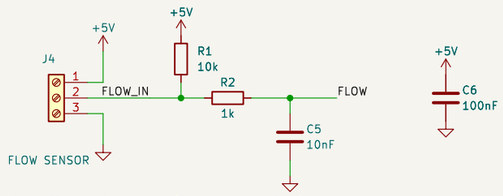
POWER: 12 V IN, FUSE, REVERSE-POLARITY CROWBAR, 5 V REGULATOR



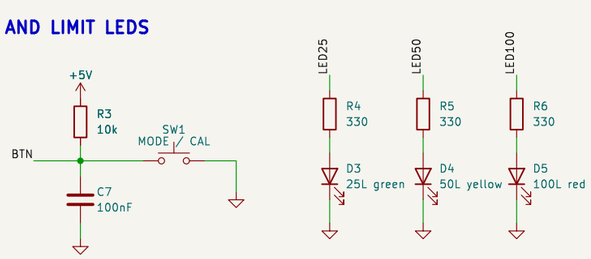
F1 + D1: if the supply is connected backwards, D1 conducts and blows F1.
U1: the ESP32-C3 Wi-Fi peaks take the 5 V load to about 0.5 A. Use a pin-compatible switching regulator (e.g. R-78E5.0-1.0) or a 5 V buck module wired to IN / GND / OUT. An L7805 needs a heatsink.

FLOW SENSOR INPUT (YF-G1, NPN open collector)

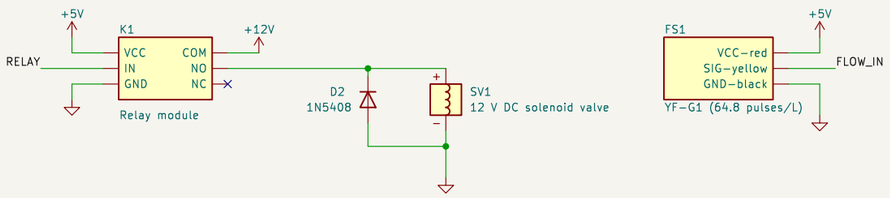


R2 + C5 filter fast spikes (time constant 10 us); the firmware also ignores pulses < 2 ms apart.
C6 sits right at J4.

BUTTON AND LIMIT LEDS

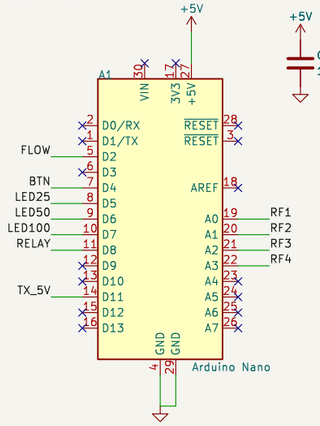


OFF-BOARD WIRING (connects to J2-J4; not placed on the PCB)



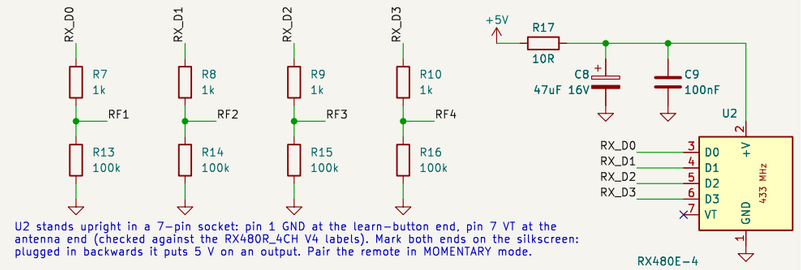
Valve circuit: relay COM <- J2 pin 1 (+12 V), valve - -> J2 pin 2.
D2 sits on the valve terminals.

CONTROLLER



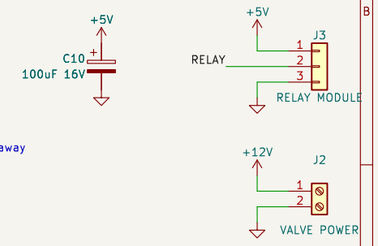
D12 is reserved for a future ESP -> Nano link.
Nano is powered through its 5V pin (VIN unused).

REMOTE RECEIVER (RX480E 433 MHz module, plugs into the board)



U2 stands upright in a 7-pin socket: pin 1 GND at the learn-button end, pin 7 VT at the antenna end (checked against the RX480R_4CH V4 labels). Mark both ends on the silkscreen: plugged in backwards it puts 5 V on an output. Pair the remote in MOMENTARY mode.

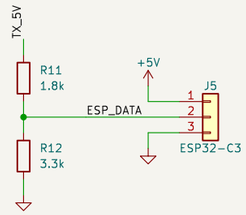
RELAY MODULE AND VALVE POWER



J2: run the valve +12 V and return as their own pair, and keep them away from the sensor cable. Place J1 and J2 side by side.
C10 sits at J3 to steady 5 V when the relay coil switches.

ESP DATA OUT (9600 baud, 3.3 V level)

ESP32-C3-Zero: J5 5V -> 5V pin, DATA -> GPIO20 (RX), GND -> GND. Its GPIOs are 3.3 V only, so R11 / R12 drop the Nano's 5 V data to 3.2 V.



MOUNTING HOLES (M3)



INTERFERENCE AND RELIABILITY MEASURES

1. D2 flyback diode right at the valve terminals (stripe to +).
2. Valve current has its own pair (J2) back to J1: star ground at J1.
3. Decoupling: C1 + C2 on 12 V, C3 + C4 at U1, C10 at J3, C11 at the Nano, C6 at J4.
4. Flow input: R1 10k pull-up, R2 1k + C5 10nF filter, firmware glitch filter.
5. Button: R3 10k + C7 100nF. Remote: R7-R10 1k series, R13-R16 100k pull-downs, receiver supply filtered by R17 + C8 + C9. Firmware debounces both.
6. F1 fuse + D1 crowbar protect against shorts and a reversed supply.
7. R11 / R12 divide the 5 V data line down to 3.2 V for the ESP.
8. Twist the flow sensor signal with its GND; route it away from the valve wires.
9. Ground plane on the bottom layer; short, wide 12 V tracks; antenna away from relay and J2.

Pins: D2 flow, D4 button, D5-D7 LEDs, D8 relay, D11 ESP, A0-A3 remote
Firmware: WaterValveController.ino
YF-G1 flow sensor, relay-switched 12 V DC solenoid valve, RX480E 433 MHz receiver, ESP data out
Arduino Nano carrier board

Sheet: /
File: WaterValveController.kicad_sch

Title: Water Valve Controller

Size: A3	Date: 2026-10-10	**Rev: 1.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1